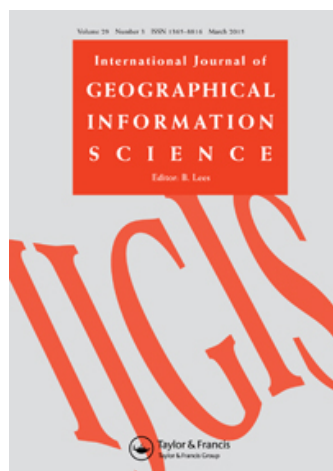


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Modelling surface drainage patterns in altered landscapes using LiDAR

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Modelling surface drainage patterns in altered landscapes using LiDAR

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The detailed topographic information contained in light detection and ranging (LiDAR) digital elevation models (DEMs) can present significant challenges for modelling surface drainage patterns. These data frequently represent anthropogenic infrastructure, such as road embankments and drainage ditches. While LiDAR DEMs can improve estimates of catchment boundaries and surface flow paths, modelling efforts are often confounded by difficulties associated with incomplete representation of infrastructure. The inability of DEMs to represent embankment underpasses (e.g. bridges, culverts) and the problems with existing automated techniques for dealing with these problems can lead to unsatisfactory results. This is often dealt with by manually modifying LiDAR DEMs to incorporate the effects of embankment underpasses. This paper presents a new DEM pre-processing algorithm for removing the artefact dams created by infrastructure in sites of embankment underpasses as well as enforcing flow along drainage ditches. The application of the new algorithm to a large LiDAR DEM of a site in Southwestern Ontario, Canada, demonstrated that the least-cost breaching method used by the algorithm could reliably enforce drainage pathways while minimizing the impact to the original DEM.

Keywords: digital elevation model; LiDAR; topographic depressions; flow-path modelling

1. Introduction

Digital elevation models (DEMs) are commonly used to simulate surface flow paths in landscapes where topography partially controls the distribution of near-surface water. Many hydro-geomorphological applications, such as stream network extraction (Jenson and Domingue 1988, Fairfield and Leymarie 1991), watershed mapping (Band 1986) and erosion modelling (Wilson and Gallant 1996), are based on deriving drainage patterns from DEMs. Extensive research has been carried out in the past to improve the algorithms used in DEM-based flow path modelling (O'Callaghan and Mark 1984, Tarboton 1997, Orlandini and Moretti 2009) and improving the quality and resolution of DEM data used in these applications (Gao 1997, Wise 2000). High-quality, fine-resolution DEMs, such as those derived from airborne light detection and ranging (LiDAR), offer new opportunities for improved accuracy in flow-path modelling.

At finer resolutions, DEM data typically contain topographic information related to the human infrastructure within a landscape. For instance, road and railway embankments and drainage ditches are often represented in DEMs of rural landscapes. Construction of

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transportation infrastructure, including road and railway embankments and their associated drainage ditches, has been shown to increase drainage densities, increasing the volume of storm flow and yielding flashier basin hydrological responses (Wemple *et al.* 1996, Tsunokawa and Hoban 1997). Anthropogenic modifications to the surface drainage network can also alter catchment boundaries and change the pre-modified flow paths (Duke *et al.* 2003, Barber and Shortridge 2005), with implications for water quality and downstream delivery of sediment (Wemple *et al.* 2001). Thus, the inclusion of anthropogenic infrastructure in fine-resolution DEMs provides the potential to improve surface drainage modelling, particularly in low-relief, rural landscapes.

Fine-resolution DEMs also present unique problems for flow-path modelling. First, DEMs represent road surfaces but do not include underpasses, such as culverts and bridges (Barber and Shortridge 2005). Underpasses are essential for properly resolving drainage patterns within these areas (Jones *et al.* 2000, Duke *et al.* 2006). Embankments in DEMs usually result in numerous artificial ‘dams’ that block flow from continuing further downstream, often creating artefact reservoirs in the topographic depressions behind the embankments (Duke *et al.* 2003, Barber and Shortridge 2005). Secondly, while open drainage ditches frequently appear alongside embankments in LiDAR DEMs of rural landscapes, they are usually incompletely represented because their widths approach the DEM grid resolution.

DEM-based flow routing applications have always had to contend with the frequent occurrence of artefact topographic depressions and flat areas within DEMs (O’Callaghan and Mark 1984). These issues are usually handled during a drainage enforcement step in which the individual elevations within the DEM are modified to ensure continuous drainage patterns (Lindsay and Creed 2005). Common approaches to drainage enforcement include topographic depression filling (e.g. Jenson and Domingue 1988, Planchon and Darboux 2002, Liu *et al.* 2009), depression breaching (e.g. Rieger 1998, Soille *et al.* 2003, Soille 2004a, 2004b) and algorithms combining both (Martz and Garbrecht 1999, Lindsay and Creed 2005). Depression filling is by far the most commonly used drainage enforcement technique. Unfortunately, these drainage enforcement techniques are poorly suited to resolving the problem of unrepresented embankment underpasses in LiDAR DEMs and can result in extensive areas of erroneous data (Barber and Shortridge 2005).

It is also common to use ancillary data, particularly vector stream networks, in the drainage enforcement process. One such technique is the practice of lowering the elevations in a DEM along stream lines, known as ‘stream burning’ (Callowa *et al.* 2007), which is then followed by depression filling to fix the remaining unresolved topographic depressions. Stream burning is appropriate when the available stream data are more accurate than the DEM. Otherwise, this technique can result in erroneous parallel streams channels, which disrupt modelled flow patterns. This is particularly the case in headwater areas where streams are often poorly mapped (Montgomery and Foufoula-Georgiou 1993). Duke *et al.* (2003, 2006) incorporated ancillary drainage information directly into the flow routing algorithm, an approach suited to application with coarse-resolution DEMs that do not represent anthropogenic infrastructure. This flow algorithm requires accurate road, drainage ditch and culvert data, which may not be widely available for some locations. Nonetheless, the work of Duke *et al.* (2003, 2006) is the most comprehensive attempt to model the impact of transportation infrastructure on regional drainage patterns in rural landscapes to date.

The difficulties in applying existing techniques to LiDAR data have meant practitioners frequently manually modify their DEMs to ensure the hydrological integrity of the data (Barber and Shortridge 2005). While this can produce good results, it is

extremely labour intensive. Schwanghart *et al.* (2013) recently demonstrated the potential for using a breaching-based algorithm to hydrological condition fine-resolution DEMs in low-relief, agricultural landscapes without the need for ancillary data or extensive manual data manipulation. This paper builds on this previous work by presenting and evaluating a new depression breaching algorithm for dealing with situations where drainage patterns cannot be resolved because of the occurrence of anthropogenic infrastructure in fine-resolution DEMs. Specifically, we assess how reliably this new algorithm can trench across embankments and other obstructions in areas of known underpasses and we compare the impact of the algorithm compared with that of the more commonly applied approaches of depression filling and stream burning drainage enforcement methods.

2. Methods

2.1. Breaching algorithm description

Depression filling raises the elevations of grid cells contained within topographic depressions to the elevation of the outlet cell. Although numerous algorithms have been developed to fill depressions within DEMs, they differ only in their approach and computational efficiency. The solution resulting from any depression filling algorithm is identical. Depression breaching by comparison lowers the grid cells along a breach channel, or trench, through an artefact dam or drainage blockage (Rieger 1998). Depression filling yields a solution that is always contained within the DEM, whereas the trench channel of a breaching operation is external to the depression feature. Therefore, with depression breaching, there are many possible solutions. A breached channel must be created in the DEM between a grid cell with no downslope neighbour (pit cell) and a distant grid cell of lower elevation (target cell). There may be many contending target cells and multiple possible breach paths between the pit and target cells. Some authors have suggested restricting the length of the breach channel (Martz and Garbrecht 1999), sometimes referred to as constrained breaching (Schwanghart *et al.* 2013).

Some breaching algorithms only consider straight trench channels (Lindsay and Creed 2005), which can result in unnatural flow patterns (Martz and Garbrecht 1999). Soille *et al.* (2003) and Soille (2004a) present an alternative breaching algorithm in which the path of the breached channel is determined by the flood order of the DEM, that is, the sequence of grid cells that become connected as the landscape is inundated from its edges moving inward and from lower cells to higher points. While this technique generally provides superior results compared with straight-line breaching, it can result in unsatisfactory flowpaths in lower-relief areas. Schwanghart *et al.* (2013), inspired by the earlier work of Soille (2004b), present another alternative breaching algorithm based on choosing a breach path that follows a least-cost optimization. The depression breaching algorithm presented in this paper is similar in that it also uses a least-cost approach to identify the breach pathway. However, the two algorithms differ in significant ways.

Regardless of the method for determining the breach path, if there is a pit cell in the DEM for which the only lower elevations occur beyond the edge of the grid, there may be no breaching solution. These depressions may be resolved using subsequent depression filling. It should be noted, however, that this final filling will have

significantly less impact of the DEM than it would had the breaching step not been used prior.

The new breaching algorithm differs from previous algorithms in the way the target cell (i.e. the lower elevation cell to which the pit will be connected) and the breach path are determined. The new breaching algorithm uses least-cost path (LCP) analysis to determine the breach trench and the target cell (Figure 1). The cost of trenching a breach channel through a particular grid cell is determined by the amount DEM elevations are lowered to ensure a monotonic downslope path connecting pit and target cells. A cost–distance determined breach trench is the pathway resulting in the least cumulative modification to the DEM for a breaching solution. The Schwanghart *et al.* (2013) algorithm, by comparison, uses an LCP approach to determine the pathway of a breach channel connecting a pit cell its sill or spillway and then used the steepest descent algorithm to connect the spillway to a downslope cell that is lower than the pit. With our algorithm, the breach path will not necessarily pass through the spillway (and thus it does not need to be identified) and the terminus is drawn from the entire set of grid cells within the extended neighbourhood that are lower than the pit cell by a specific, distance-related value.

There are several advantages to the cost–distance approach to depression breaching, particularly where there are transportation embankments. Where there is a pit cell within a roadside ditch, the algorithm can trench a path to a more distant location along the ditch

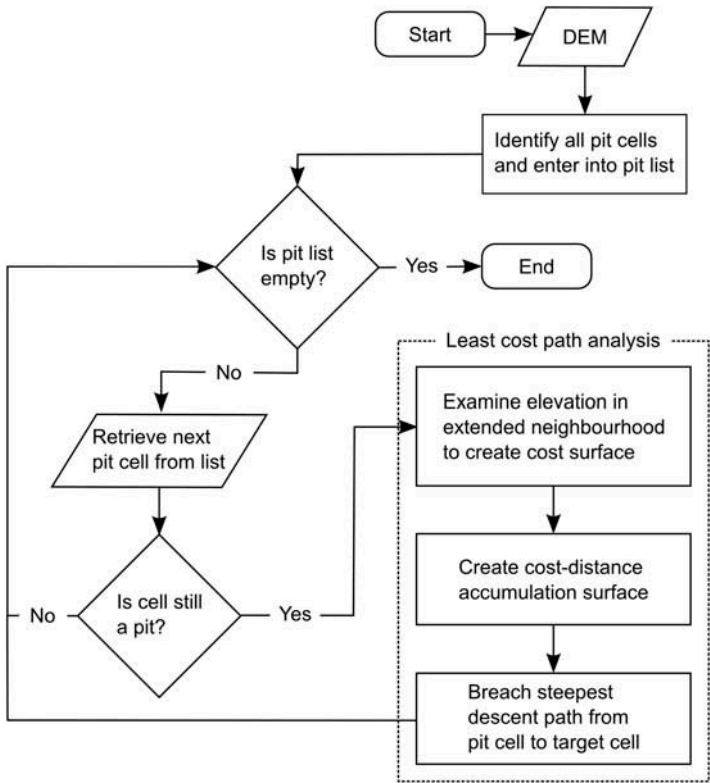


Figure 1. The least-cost-based breaching algorithm.

rather than breaching to a nearer target cell on the other side of the road. This serves to enforce drainage along ditches, which is less costly than breaching through embankments. On the upstream side of culverts passing through a transportation embankment, the algorithm will breach a trench through the embankment in the location of the culvert because the only target cells are downstream of the road/railway. Thus, breaching provides a good means of both improving drainage along ditches and coping with the artefact dams created when streams pass through culverts. Furthermore, this is achieved without the need for ditch or culvert data.

Pit cells also frequently occur along narrow stream channels for the same reason they are common along drainage ditches, that is, channel widths approach the grid resolution. Other depression breaching algorithms can straighten the stream course in these situations, a poor solution for meandering streams. The new breaching algorithm, however, can enforce drainage along the original stream course rather than artificially straightening the channel. Similarly, the cost–distance approach to breaching can enforce natural drainage pathways on hillslopes where pits are abundant.

The algorithm starts by finding each cell in the grid that has no lower neighbouring cell and entering these pits into a list. The list is then sorted in ascending order of elevation. The first cell in the list is visited and the elevations within an extended neighbourhood centred on the pit cell are examined. The neighbourhood is of a user-specified dimension, related to the maximum allowable breach distance ($d_{\max}$). Cost values (C) are calculated for each grid cell in the neighbourhood by differencing each cell's elevation and the pit's elevation. Grid cells that have elevations lower than the pit are potential targets and have no associated cost. The C -values are stored in memory in a two-dimensional array. This cost surface grid serves as the input for the LCP analysis. The breaching algorithm performs an LCP analysis for each grid cells in the pit list.

Large LiDAR DEMs often contain millions of pit cells. Therefore, a highly efficient LCP method was used in the algorithm implementation. The LCP analysis begins by creating a cost–distance accumulation surface from the input cost surface (Figure 2). The cost–distance accumulation value (A) in each grid cell is the minimum accumulated cost of travelling from the cell to a target cell. The accumulated cost of travelling from cell (i, j) to its neighbouring cell (m, n) can be calculated as

$$A_{m,n} = A_{i,j} + \Gamma(C_{i,j} + C_{m,n})/2 \quad (1)$$

where Γ is the distance between the cells, taking into the account the greater distance between diagonal neighbours.

Once the cost–distance accumulation surface is calculated, the breach trench can be determined by following the steepest descent pathway over this A -surface. This pathway begins at the pit and ends at the target cell. Most of the computational effort therefore is spent in calculating the cost–distance accumulation surface (Figure 2). Elevations in the output DEM along steepest descent pathway initiated at the pit cell and terminating the first-encountered target cell are decremented by the C -value calculated earlier in the process.

The algorithm rejects potential breach paths that require decrementing elevations in the DEM greater than a user-specified height. The breaching algorithm was implemented as a plug-in tool within the open-source software Whitebox Geospatial Analysis Tools. The source code

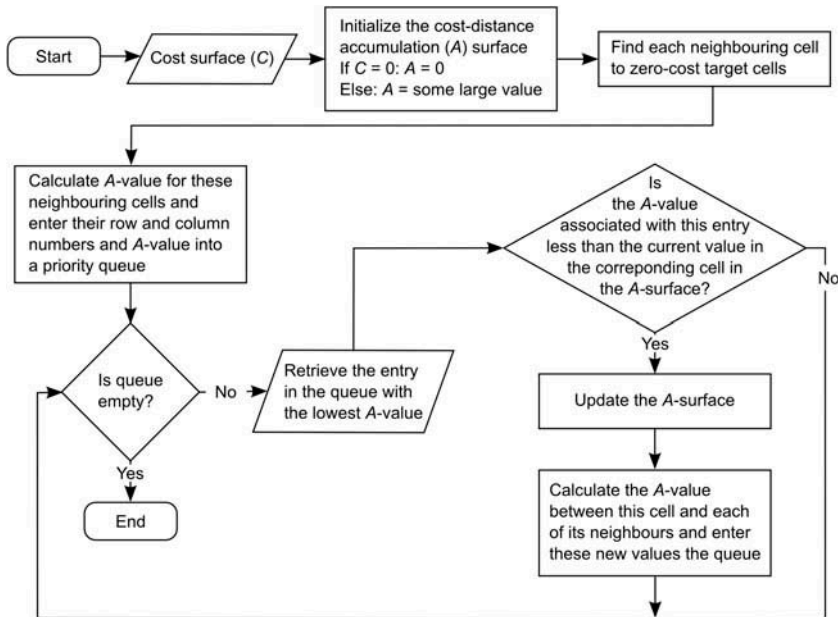


Figure 2. The cost-distance accumulation process used in the breaching algorithm.

can be downloaded from the project webpage at <http://www.uoguelph.ca/~hydrogeo/Whitebox/>. The algorithm has been designed to work with very large (>1 GB) DEMs.

2.2. Study site and DEM data

Algorithm performance was evaluated for the Rondeau Bay area, located in Southwestern Ontario, Canada, along the coast of Lake Erie (Figure 3). The Blenheim moraine transects the study site. The Thames River drains the region north of the moraine towards Lake St. Clair. The area south of the moraine drains into Rondeau Bay through incised stream channels. The morainal ridge rises approximately 20 m above the surrounding flat former lakebeds. Drainage systems on the moraine are poorly organized owing to the undulating topography and ditches are used to improve drainage. Soils include gravelly loam with moderately high organic matter on the moraine and sandy loam and silty clay loam covering much of the remaining areas (Walker and Richards 1994). Agriculture is the dominant land use and urban areas comprise less than 8% of the site, including the towns of Chatham, Blenheim and Ridgeway.

A commercial data provider collected LiDAR data of the site over a 2-day period in early May 2008. The flight occurred during optimal weather conditions and during the leaf-off season. The average point density was 1.3 points m^{-2} . The raw LiDAR point data, a collection of 586 1 km^2 tiles, were interpolated into a 28 219 $\times$ 33 380 pixel, 1 m resolution DEM using the inverse-distance weighted algorithm in the Whitebox GAT software. The final grid, composed of the mosaicked 1 km^2 interpolated tiles, contained extensive areas of NoData values surrounding the irregular-shaped study area. Only last return points were used in the interpolation. Off-terrain

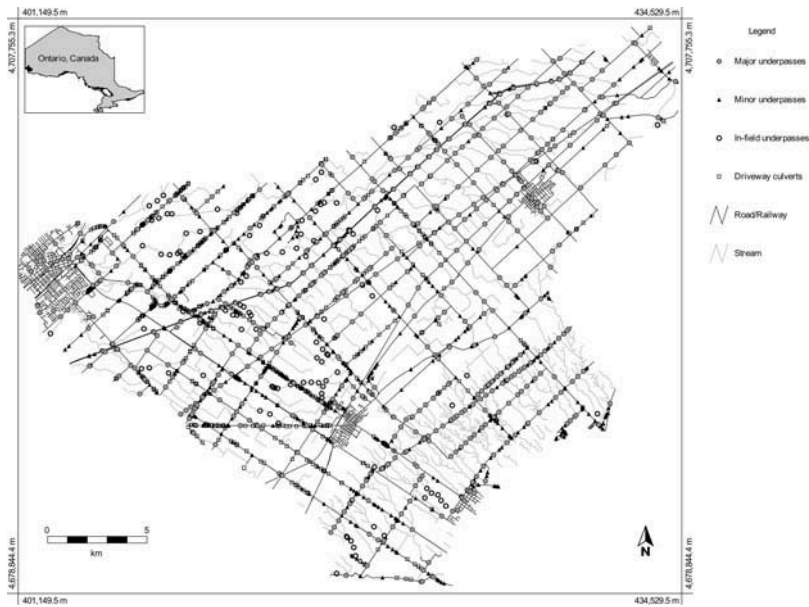


Figure 3. Mapped underpasses within the Rondeau Bay study site.

objects, including buildings and vegetation, were removed from the elevation model to create a bare-earth DEM.

2.3. Reference data

A 0.30 m resolution, colour air photo data set was used for mapping underpasses. Culverts could be identified by their concrete and steel sections extending beyond the embankment, as well as by their adjacent plunge pools. Vector road and railway networks provided additional information for manual mapping. Urban areas were excluded from the analysis.

In total, 2029 underpasses were identified in the Rondeau Bay site (Figure 3). Of these mapped underpasses, 577 (28.4%) were major drainage features, serving as conduits for perennial streams. A similar number (590) were small diameter culverts that could not transfer substantial quantities of water. These features, often connecting roadside ditches on either side of the road embankment, were classified as minor embankment-crossing underpasses. The largest category of mapped underpasses (758) included the culverts along roadside ditches associated with driveways. Although relatively small diameter culverts, these features were largely responsible for interrupting flow paths along drainage ditches within the LiDAR DEM. An additional 104 underpasses were categorized as in-field underpasses, consisting of footbridges in parks or culverted sections of streams within agricultural fields. Field visits were used to corroborate the data set and to provide clarification for ambiguous cases. A sample of 440 mapped underpasses was verified in the field in the autumn of 2010. A small number of underpasses were mapped during the field visit, that is, features not previously identified in the imagery, usually because off-terrain objects and shadows in the imagery obscured them.

2.4. Test data

The breaching algorithm was applied to the study site DEM. The selection of the maximum allowable breach distance ($d_{\max}$) parameter required careful consideration. There are few consequences to setting $d_{\max}$ to a large value ensuring a solution for worst-case scenario pits with very distant targets. However, any LiDAR DEM of comparable size to the study site data set (3.5 GB) is likely to contain millions of pits, the majority of which would be located on hillslopes and associated with small-scale roughness and white noise. These pits frequently have target cells within relatively short distances. It is inefficient to perform the LCP analysis for a large neighbourhood extending past the target cell but it is impossible to know the location of the target cell *a priori*. Therefore, setting $d_{\max}$ to a small value greatly reduces processing time. If $d_{\max}$ is too small, however, for some pits a nearer, higher-cost solution may be preferred over an undiscovered lower cost but longer trench. A good compromise is to run the breaching algorithm multiple times, increasing $d_{\max}$ with each run, and feeding the output DEM from the previous run into the subsequent run. The algorithm was run for the Rondeau Bay DEM using $d_{\max}$ values of 5, 50, 150, 750 and 1500 m, respectively. A maximum allowable decrement height of 10 m was used in each run.

There are usually pits for which a breach solution cannot be found because no appropriate target cell can be located. This results where (1) the target is beyond the edge of the data set, (2) the search distance is too small to locate the target and (3) the available solutions require a trench of an unacceptable depth. These final pit cells can be resolved using a depression filling algorithm, creating a hydrologically conditioned DEM with continuous flow paths. The $d_{\max} = 1500$ m DEM was processed using a filling algorithm, producing the final breached DEM.

3. Results

Table 1 shows the results of applying the breaching algorithm to the Rondeau Bay DEM. The processing time for each step in the breaching series was more closely related to the $d_{\max}$ value than to the number of depressions processed in the step. The overall processing time was nearly 23 hours running on a computer with a quad-core 2.7 GHz processor and 16 GB of memory. The data set was found to contain 3,776,252 pit cells, of which 99.95% could be resolved using the breaching algorithm, leaving 1766 pit cells that were subsequently filled. Most pits (76.70%) were solved using a relatively short breach channel of approximately five cells (Table 1).

Table 1. Results of applying the depression breaching algorithm multiple times to the Rondeau Bay DEM, with increasing $d_{\max}$ value, measured in grid cells.

$d_{\max}$	Solved pits	Pits remaining	Processing time
5	2,896,426	879,826	00:03:25
50	811,649	68,177	01:21:59
150	49,448	18,729	06:13:01
750	14,593	4136	09:43:45
1500	2370	1766	05:27:27
Overall	3,774,486	1766	22:49:37

3.1. Impact of the depression breaching algorithm on the DEM

A pre-processing algorithm's impact can be assessed by differencing the modified and original DEMs. Figure 4 shows the impact of the breaching algorithm compared to the filling and stream-burning solutions. The impact of breaching was minor for most of the study area (Figure 4a). Most higher impact areas were associated with the 1766 unbreachable pits that were resolved by depression filling. This included a large area near Rondeau Bay for which the breaching algorithm was unable to find solutions because there were no lower cells behind high road embankments along the coastline. The other high-impact area was along lower reaches of the Thames River in the north of the study site where a water diversion caused breaching difficulties. The breaching algorithm also created deep trenches at sites of road underpasses, as expected. Figure 5, which shows the impact of breaching at an embankment trench site, also illustrates enforced flow along the incised stream channel and several driveway culverts. The breaching algorithm resulted in a volumetric modification to the DEM of 6,224,882.2 m³.

The filling algorithm impacted the DEM more extensively than depression breaching (Figure 4b). Along the incised streams of the southern portion of the study site, filling flooded the incised valleys upstream of embankments. Depression filling raises the DEM in these depressions, replacing the natural topography with a gently sloping surface draining towards the outlet along the embankment. All of the surface drainage information within flooded areas was lost. This impact was not restricted to incised valleys, however. Figure 4b shows that extensive areas of agricultural fields upstream of embankments were also affected by this 'digital flooding'. Although many of these areas were only slightly elevated, all of the information about drainage contained in the original DEM was lost through depression filling. The breaching algorithm was found to modify the elevations of 86.5% fewer grid cells than the depression filling solution. Depression filling also modified a considerably larger volume of 55,923,147.8 m³.

Figure 4c shows the impact to the DEM that results from applying stream burning to the DEM prior to depression filling. While stream burning eliminated much of the impact caused by filling within valleys by trenching across embankments, it did not reduce the impact within fields upstream of embankments. Depression breaching modified 81.2% fewer grid cells than the stream burning solution. While the modified volume resulting from stream burning was less than that of depression filling, at 33,766,908.1 m³ it was still substantially greater than the modification resulting from the breaching solution.

3.2. Breaching underpasses and enforcing flow along drainage ditches

Of the 2029 mapped underpasses, 79.3% intersected or were near trenches (Table 2). For the purpose of the analysis, all underpasses within 12 m of trenches were considered successfully mapped. There was a large difference in the breaching accuracy of major and minor embankment-crossing underpasses. The algorithm breached the embankments of 87.7% of the underpass sites classified as major embankment-crossing drainage features, while only 61.0% of minor underpasses were breached. All of the in-field underpasses were breached (Table 2). The algorithm successfully breached 84.4% of the 759 along-ditch culverts buried beneath driveways, which indicates the algorithm's ability to enforce drainage along roadside ditches.

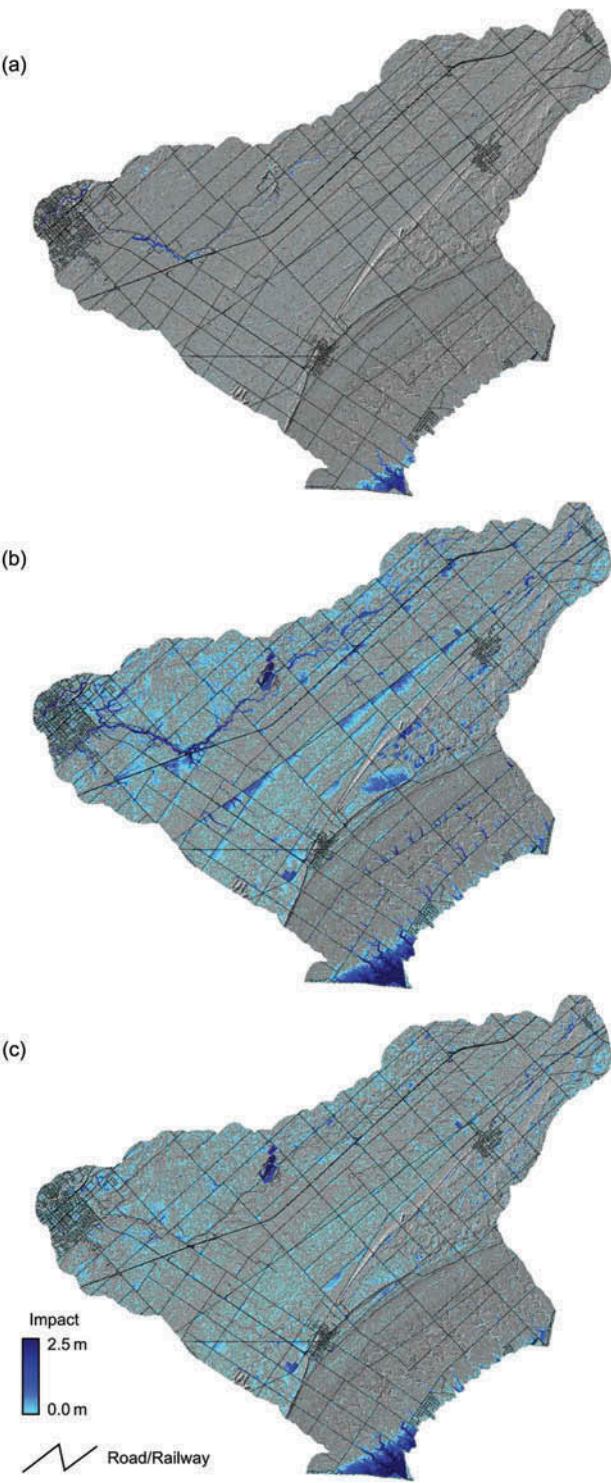


Figure 4. The impact of (a) depression breaching, (b) depression filling and (c) stream burning on the LiDAR DEM. Impact is the absolute elevation difference between the original and processed DEMs. Impact maps overlay grey-tone shaded-relief images.

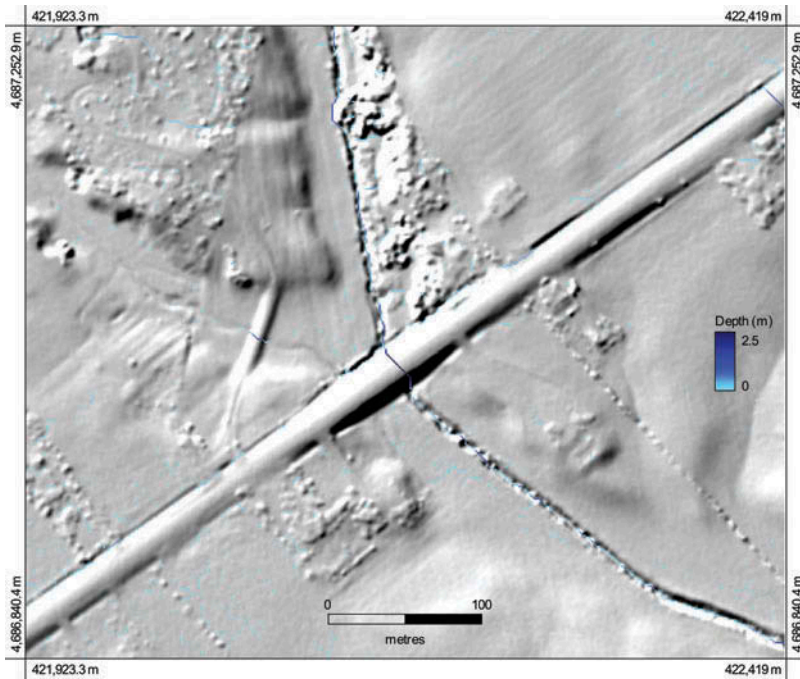


Figure 5. Detailed image of the breaching impact for a major road underpass.

Table 2. Accuracy assessment of the depression breaching algorithm for correctly identifying the location of various underpass types in the Rondeau Bay study site.

Underpass type	Number	Per cent breached
Major	577	87.7
Minor	590	61.0
In-field	104	100.0
Driveway	758	84.4
Overall	2029	79.3

4. Discussion

All unprocessed DEMs contain pits, which occur either because of error or the inability of a surface model to represent actual flow paths, as is the case when a DEM contains transportation embankments. The fine resolution and speckle-type error associated with LiDAR data make these DEMs particularly vulnerable to both types of pits. Any method for resolving flow paths for pits will have unintended consequences for other grid cells, resulting in loss of surface drainage information. The findings showed that the breaching algorithm was able to resolve flow paths for most pits in the DEM while requiring substantially less modification than depression filling or the stream-burning method.

The least-cost approach to depression breaching provides several theoretical advantages over alternative breaching techniques, particularly for resolving flow paths in DEMs containing transportation embankments and streams. The algorithm will inherently prefer solutions that require less modification to the DEM. Yet it is flexible enough to identify longer but lower-cost breach channels. This characteristic allows the algorithm to enforce flow along channels that are partially blocked in the DEM, for example, along roadside ditches, canals and incised rivers. Because the least-cost breach channels can curve, this method works well along rivers, where other breaching methods can produce cut-offs leading to channel straightening. In locations where streams pass beneath embankments, either within culverts or under bridges, the algorithm should breach through embankments, connecting the upstream and downstream sides of the feature.

The Rondeau Bay study site analysis provided an opportunity to evaluate how well these theoretical advantages translated to a real-world application. This analysis demonstrated that the algorithm is capable of being used with massive DEMs, consisting of at least one billion cells. The processing time for the breach algorithm is however greater than that of modern depression filling algorithms. Much of the computation is used in the LCP analysis, which was strongly affected by the search distance parameter. Because most of the pit cells in any LiDAR DEM will possess short-distance solutions (Table 1), it was deemed acceptable to iteratively increase the search distance. While this made processing feasible, it may have affected the final result for some pits.

The algorithm performed very well at breaching embankments at sites where streams crossed roads and railways. Furthermore, obstructions to flow paths along rivers and drainage canals within the DEM were handled well. As expected, the curved breach paths produced by the algorithm were found to enforce natural channel pathways for the meandering sections of rivers. Overall, the breaching solution was superior to stream burning. The mapped stream network available for the study site frequently deviated from the path of the stream within the DEM, resulting in numerous sections of parallel channels in the stream-burned DEM. The vector streams were of a poorer resolution and suffered cartographic generalization errors. Therefore, the LiDAR DEM represented streams more accurately overall.

Open ditches usually parallel embankments in rural areas. These features serve to drain the paved surface and the surrounding areas and to ensure the embankment does not become saturated or inundated. Examination of the breach channels within the LiDAR DEM (Figure 4a) did reveal substantial lengths of breached channels along roadside ditches throughout the study site. The algorithm often breached driveway culverts (Table 2). However, the enforcement of flow paths along roadside ditches did provide some problems for the breaching algorithm. Minor culverts were breached less reliably (Table 2). Furthermore, the algorithm was found to trench across embankments in locations where there was no evidence for an underpass. This liberal embankment breaching was more prevalent along roads with lower gradients, that is, within ditches with very gentle along-ditch gradients. Therefore, in some cases it appears the algorithm could not identify potential target cells within the search distance along gently sloped roadside ditches, and instead found target cells on the opposite side of the embankment.

This error may have resulted because the distance parameter was increased in steps. Improvements in algorithm computational efficiency may allow for the use of initially longer search distances, thereby minimizing this effect. However, drainage ditches are often more gently sloped than natural channels, sometimes resulting in ambiguous flow directions. Thus, this observed inappropriate trenching across embankments along flat roadside ditches might be an inherent limitation of the breaching method. Nonetheless,

this situation is an improvement over the depression filling solution, which floods the upstream area behind an embankment (Figure 4b), providing pathways for water to be transmitted over the feature along its entire length, effectively ignoring the effect of drainage ditches all together.

The breaching solution presented in this paper is capable of substantially improving the representation of surface hydrological flow paths within a DEM without relying on any ancillary information such as drainage ditch and stream vectors. This is particularly advantageous when the DEM is of a higher resolution than the drainage information, which is often the case when working with highly detailed LiDAR DEMs.

Lindsay and Creed (2006) advocate maintaining real topographic depressions in DEMs while selectively removing artefact depressions. They also describe techniques that can be used to distinguish real depression features from artefacts. The study site in this paper did contain actual depressions, many of which had been effectively removed from the landscape through construction of drainage ditches. Nonetheless, if a set of real depressions were identified in advance using one of the techniques described by Lindsay and Creed (2006), it would be possible, if warranted by the application, to modify the breaching algorithm to remove the associated pit cells from the pit cell list prior to the onset of the breaching process, thereby maintaining these features in the DEM.

5. Conclusions

This paper presents a new breaching algorithm for enforcing flow paths in LiDAR DEMs. The algorithm possesses advantages over existing methods, particularly for applications in human-modified landscapes containing road and railway embankments and drainage ditches. The performance of the algorithm was evaluated using a large LiDAR data set of a rural landscape in Southwestern Ontario. The following conclusions were drawn from this work:

- (1) The breaching algorithm generally performed well at enforcing flow paths along drainage features such as streams and drainage canals without the need for ancillary data.
- (2) Bridges and major culverts were handled by trenching a path connecting the upstream and downstream sides of road/railway embankments in 87.7% of the cases.
- (3) While the breaching algorithm was found to enforce flow along roadside ditches over much of the study site, ditches of low slope gradient did experience some erroneous trenching across the embankment. This may have resulted from the use of an initially low search distance parameter; however, for ditches with gentle gradients, this characteristic may be inherent in the approach.
- (4) The least-cost breaching method was found to impact the DEM by at least 80% less than either depression filling or stream-burning with subsequent filling. Thus, the breaching algorithm was able to take advantage of more of the surface drainage information contained in the original DEM.

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